De La Salle University - Manila

Silicon Minds: The Evolution of the Mobile Microprocessor

Virtual Exhibit Proposal

In Partial Fulfillment of the

Course Requirements for

INTRODUCTION TO COMPUTER ORGANIZATION AND ARCHITECTURE 2 (CSARCH2)

Term 3 A.Y. 2025-2026

1. Group Information

1.1 Proposed Title

Silicon Minds: The Evolution of the Mobile Microprocessor

1.2 Member Roster

#	Full Name	GitHub Handle
1	MARIA FIDES BANCORO	@m-fides
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1.3 Detailed Project Description

Silicon Minds: The Evolution of Mobile Microprocessor is an immersive, interactive virtual museum tracing the architectural journey of the mobile microprocessor. From their origins as simple processors designed for mobile communication devices to their current role as the engines powering smartphones, tablets, and other everyday mobile devices, the museum explores how mobile processors have evolved and improved over the years to meet modern computing demands.

Visitors are guided through five landmark eras in mobile microprocessor development, namely: The Birth of Mobile CPUs, the ARM Revolution, the Multicore Era, the System-on-Chip Era, and the AI & Efficiency Era. Each era is framed around a compelling narrative arc of the limitations that drove innovation, the architectural breakthroughs that followed, and the ripple effects these advancements had on mobile device performance, power efficiency, and user experience. Rather than presenting history as a sequence of technical specifications, the exhibit tells the story of how the constantly increasing demands for speed, battery life, and multitasking influenced the design of mobile microprocessors.

To make these concepts tangible, the exhibit integrates three core interactive experiences. The Microprocessor Timeline Explorer serves as the center of the exhibit, allowing users to explore major mobile microprocessor generations and examine their specifications. The Mobile Processor Evolution Simulator will help users visually analyze how processors from different generations handle the same workloads, revealing the improvements in performance. Lastly, the Mobile Microprocessor Trivia Quiz will test the visitors on their understanding of the presented key concepts in an engaging, low-stakes format.

At the end of the exhibit, visitors would have developed a clear, intuitive understanding of not only how mobile microprocessors evolved, but also why the innovations that emerged through the years led to the powerful processors modern mobile devices have. They will gain insight into how advancements improved and transformed devices from simple communication tools to powerful and heavily functional computing platforms used in everyday life today.

2. Topic Theme

2.1 Overview

The mobile microprocessor stands as one of the most consequential innovations in computing history. Beginning as a modest, low-power adaptation of desktop RISC designs, it has grown into a highly integrated, AI-capable system-on-chip that fits in the palm of a hand, yet rivals laptop-class performance. This exhibit traces that evolution across five distinct eras, each defined by a breakthrough in architecture, manufacturing, or capability that permanently changed what mobile devices could do. The narrative is organized around the engineering tensions that drove each transition — from battery-life constraints that forced creative core design to consumer demand for richer applications that pushed transistor counts into the billions.

2.2 Five Eras of GPU Evolution

Era	Period	Key Milestone
Birth of Mobile CPUs	Mid-1990s–Early 2000s	Early low-power RISC processors adapted for mobile phones. E.g., ARM7TDMI, Intel StrongARM, Motorola DragonBall.
The ARM Revolution	2007–2012	ARM architecture dominates smartphones; dedicated ISPs and DSPs emerge. E.g., ARM Cortex-A8, Apple A4, Qualcomm Snapdragon S1.

The Multicore Era	2011–2014	Dual- and quad-core mobile CPUs arrive, boosting multitasking and app performance. E.g., NVIDIA Tegra 2, Apple A6, Qualcomm Snapdragon 600.
The System-on-Chip Era	2015–2020	Highly integrated SoCs combine CPU, GPU, modem, and memory on a single die. big.LITTLE heterogeneous cores improve power efficiency. E.g., Apple A9, Qualcomm Snapdragon 820, Samsung Exynos 8890.
The AI & Efficiency Era	2021–Present	Dedicated NPUs enable on-device AI; TSMC 3nm/4nm nodes push performance-per-watt to new heights. E.g., Apple A17 Pro, Qualcomm Snapdragon 8 Gen 3, MediaTek Dimensity 9300.

3. Tech Stack Plan

3.1 Core Technologies

The following table summarizes the full technology stack, aligned to the project requirements (Node.js 26, Astro 6) and optimized for seamless integration into the central museum website.

Layer	Technology	Notes
Runtime	Node.js 26	Required per project specs; LTS stability.
Framework	Astro 6	Static-site generation with island architecture; forked from provided template.
Content	.mdx files	MDX (Markdown + JSX) for rich, component-embedded content pages.
Interactive UI	React 18 (.jsx)	All four interactive components built as React islands embedded in .mdx.
Styling	Tailwind CSS v4	Utility-first CSS; follows the central museum template style guide.
Version Control	Git / GitHub	Repository forked from template; all docs, plans, and feedback tracked via commits.
Responsive	CSS Grid / Flexbox	Mobile-first breakpoints; timeline scrolls horizontally on desktop, vertically on mobile.

Animation	Framer Motion	
Icons	Lucide React	Consistent icon system for UI controls, era tags, and tooltip pins.
Hosting	GitHub Pages	Static output deployed via astro build; automated via GitHub Actions on push to main.

3.2 Mobile Responsiveness

All components are built mobile-first:

- **Mobile Microprocessor Trivia Quiz**: Question cards stack vertically and fill the screen width on mobile (<640px). Answer buttons expand to full width for easy tap targets (min-height: 48px). Score card collapses to a single centered summary panel.
- **Microprocessor Timeline Explorer**: Horizontal scroll on desktop (≥768px); switches to a vertical stacked card layout on mobile. Expanded spec cards go full-width. Era filter buttons collapse into a horizontally scrollable pill row.
- **Mobile Processor Evolution Simulator:** On mobile (<640px), selected era panels are stacked vertically so each simulation is visible on the screen. The Era Highlights (information) and Performance Metrics sections are collapsible and can be expanded when the user chooses to do so. When switched to desktop, simulation panels are displayed side by side, while Era Highlights and Performance Metrics panels remain visible below the simulation for easy comparison.

4. Interactive Components

#	Component	Type	Summary
1	**Mobile Microprocessor Trivia Quiz**	**Quiz**	It serves as a knowledge check designed to support what users have learned after browsing through the five eras of mobile microprocessor evolution. Each question is tagged to a specific era so users know what concept is being tested. Incorrect responses reveal a short explanation clarifying the right answer.

2	**Microprocessor Timeline Explorer**	**Interactive Timeline**	The microprocessor timeline explorer, being the central feature of the museum, allows users to trace the evolution of mobile microprocessors across different eras. It will present the key mobile microprocessor generations, examine their specifications, and differentiate developments across eras.
3	**Mobile Processor Evolution Simulator**	**Simulation**	Interactive simulator that helps users visualize the evolution of mobile microprocessors by allowing them to run the same task workload across processors from different eras. Users can compare microprocessors' performances through simple real-time workload animations that will demonstrate how each generation's architectural innovations significantly improved processing speed, efficiency, and task management over time.

4.1 Mobile Microprocessor Evolution Trivia Quiz

Purpose:

- To test the user's understanding after browsing the exhibit

Interaction Details

- **Era Tagged questions:** Each question is labeled with its microprocessor era so users know what concept is being tested.
- **Multiple choice format:** 4 choices per question, and wrong answers show a short explanation why.
- **Score Card:** Shows how many per era the users got right which could motivate them to revisit their weak spots in the exhibit.
- **Randomized Question Pool:** The bank contains 30 questions (6 per era). Each session presents 15 randomly selected questions (3 per era minimum guaranteed) shuffled on every load. Answer options within each question are also randomized.

4.2 Microprocessor Timeline Explorer

Purpose:

- To allow users to visually trace the progression of mobile microprocessor architectures across all five eras, giving them a concrete sense of how dramatically performance, efficiency, and capability have evolved over the decades.

Interaction Details

- **Era Nodes:** Each landmark mobile microprocessor generation is represented as a clickable node along the timeline. Clicking a node expands a spec card displaying transistor count, core configuration, speed, efficiency improvements, and other notable architectural features.
- **Era Filter Buttons:** Users can filter the timeline to focus on a single era, dimming all other nodes to reduce visual clutter and highlight one generation at a time.
- **Spec Card Animations**: Expanded cards appear with entrance animations (500ms, cubic-bezier ease-out) consistent with the style guide, and collapse smoothly when another node is selected or the card is dismissed.
- **Comparative View:** Users can select two nodes simultaneously to display their spec cards side by side, making cross-era comparisons immediately readable.
- **Responsive Layout:** On desktop (≥768px), the timeline scrolls horizontally with nodes laid out left to right. On mobile (<640px), the layout switches to a vertical stacked card view for easier single-thumb navigation.

4.3 Mobile Microprocessor Evolution Simulator

Purpose:

- To help users better visualize and compare the performance of mobile microprocessors across eras through a simple simulation of how the different processors handle identical workloads. After the selection of processor eras, users can observe how processing speed, power efficiency, and task management capabilities differed across generations.

Interaction Details

- **Era Selection:** Before simulation starts, users would be given the option to select two or more mobile microprocessor eras to compare performance. These selected eras will serve as the basis for the side-by-side comparison.
- **Identical Workload Simulation:** All selected eras are given a standardized workload and each will simulate how their microprocessors would handle the task using visuals. Simple animations will show differences in completion time, smoothness, and efficiency of each microprocessor based on predefined processor behavior.
- **Comparative view:** The simulation will be displayed in a side-by-side layout so the users can see and compare the performances handling the same workload in real-time.

- **Microprocessor Highlights:** Simulation will also have a panel beside it that displays the important information of the microprocessor introduced on the chosen eras. This can include its name, core count, speed, and significant architecture innovations.
- **Performance Metrics and Feedback panel:** After the simulation, performance metrics will be displayed to show how each era handled the task differently in terms of processing speed, power efficiency, and etc. This feedback panel can summarize the results of the simulation so that users can better compare the microprocessors.
- **Replay Simulation Option:** After the simulation, users are given the option to restart the simulation or to select different processor eras to compare.

5. Style Guide Snapshot

5.1 Design Philosophy

The exhibit uses a high-contrast dark palette to evoke the aesthetic of system diagnostics, chip datasheets, and performance monitoring interfaces, which the visual language engineers use when working with the hardware itself. The design was sort of inspired by that of VSCode and what would be seen in futuristic games like Cyberpunk 2077. Monospace typography for technical data, layered surface cards, and a structured grid system reference the precision and density of silicon design. This creates a focused, professional museum atmosphere that lets each generational leap in mobile processor capability speak for itself, without unnecessary decoration. All color and spacing decisions below are finalized and serve as the source of truth for all component implementations.

5.2 Style Tokens

Element	Value	Usage
Primary Color	#00E5FF / cyan-400	Era highlights, active states, key callouts
Secondary Color	#7B61FF / violet-500	AI & Efficiency era accent, interactive hover, quiz feedback
Background	#0A0A0F	Page-level background, deep space feel
Surface Cards	#141420	Era cards, quiz panels, viewer container
Primary Text	#F0F0F5	Body copy, headings, labels on dark surfaces

Secondary Text	#8888A0	Metadata, dates, captions, disabled states
Typography (Headings)	Space Grotesk 500/700	Era titles, section headers, component names; geometric sans to echo silicon aesthetics
Typography (Body)	Inter 400/500, 16px / 1.7	Exhibit copy, descriptions, quiz questions
Typography (Technical Data)	JetBrains Mono 400, 13px	Spec numbers, benchmark values, table data, transistor counts, clock speed figures
Border Radius	sm: 4px md: 8px lg: 16px	sm for tags/chips md for cards, inputs lg for major panels, simulator container
Grid System	12-col, 24px gutter, 40px margin	Timeline: full-width scroll, components: 6-col or 12-col spans on desktop
Spacing Scale	4 · 8 · 16 · 24 · 40 · 64 px	4px intra-element, 16–24px intra-card, 40–64px between major sections
Shadows	0 0 0 1px rgba(0,229,255,0.12)	Subtle cyan glow-border on cards; no traditional drop shadows on dark bg
Animation Duration	fast: 150ms standard: 300ms entrance: 500ms	fast -> hover feedback; standard ->transitions; entrance -> era card reveal, quiz score
Animation Easing	cubic-bezier(0.16, 1, 0.3, 1)	Snappy ease-out for all transitions; conveys precision and responsiveness
Component Borders	1px solid rgba(255,255,255,0.08)	Default card borders; inactive timeline nodes
Interactive Feedback	Cyan glow + scale(1.02)	Hover on era cards, quiz options, simulator panels; correct answer -> green glow; wrong -> red flash
Timeline Accent	Gray -> Cyan gradient	Horizontal timeline track, past eras desaturate, current/active era glows cyan
Responsive Breakpoints	mobile: <640px tablet: 640–1024px desktop: >=1024px	Matches Tailwind sm/md/lg, timeline switches axis at 768px per spec

5.3 Accessibility

The exhibit targets WCAG 2.1 AA compliance throughout. The following requirements are binding for all component implementations:

- **Contrast Ratios:** Primary text (#F0F0F5) on background (#0A0A0F) yields 16.9:1, well above the 4.5:1 AA minimum. Secondary text (#8888A0) on background achieves 5.1:1.
- **Focus States:** All interactive elements have a visible 2px solid #00E5FF focus ring with 2px offset, replacing the browser default.
- **ARIA Labels:** All icon-only buttons carry descriptive aria-label attributes. The simulator component exposes role="region" with a text alternative summarizing the currently selected processor era and workload result.
- **Reduced Motion:** All animations respect prefers-reduced-motion: reduce — entrance animations and transitions are disabled, replaced with instant state changes.
- **Keyboard Navigation:** The timeline, quiz, and era navigator are fully operable via keyboard. The simulator supports Tab-key navigation between era panels and Enter to trigger or restart the simulation.

5.4 Style Guide Mockup

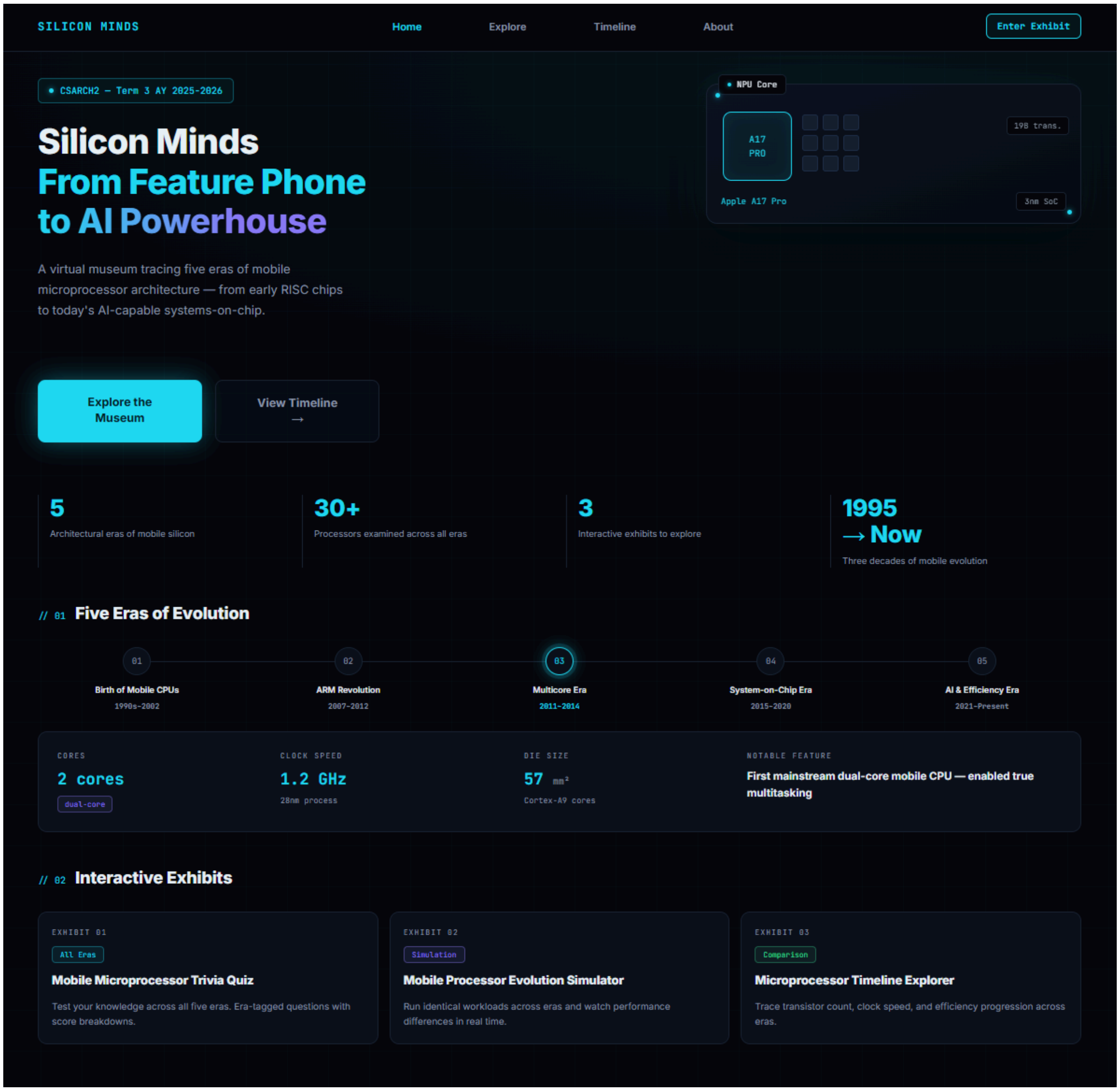

Figure 1. *Desktop Layout Mockup of Virtual Exhibit Landing Page*

Figure 2. Mobile Layout Mockup of Virtual Exhibit Landing Page